EgoTECH

SOFTWARE PROJECT PROPOSAL

EgoTECH PDF Vault — Document & Image Storage System

Document Type	Software Project Proposal (SPP)
Project Name	EgoTECH PDF Vault
Organization	EgoTECH
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1. Executive Summary

EgoTECH PDF Vault is a full-stack, web-based document management platform designed to allow individuals and small organizations to upload, organize, search, download, and delete PDF documents and image assets in a secure, multi-user environment. The system is developed by EgoTECH and is intended to address the growing need for lightweight, self-hosted document storage solutions that do not require expensive cloud subscriptions.

The platform consists of a Python/FastAPI REST backend, a React/Vite frontend, and an SQLite relational database. Users authenticate via JSON Web Tokens (JWT), manage personal folder hierarchies, and access their files through a clean, modern browser interface.

2. Problem Statement

Organizations and individuals frequently face the following document-management challenges:

- Difficulty keeping PDF documents organized across multiple devices and locations.
- Reliance on expensive, feature-heavy cloud services (e.g., Google Drive, Dropbox) for basic storage needs.
- Lack of a lightweight self-hosted solution with built-in user authentication.
- Absence of a unified workspace for both PDF documents and image assets.

EgoTECH PDF Vault directly addresses each of these pain points with a focused, easy-to-deploy application.

3. Proposed Solution

The proposed solution is a web application with two workspaces — a Documents workspace and an Images workspace — accessible after secure user registration and login.

3.1 Key Features

- User registration and login using hashed passwords and JWT bearer tokens.
- PDF upload with server-side validation (magic-byte check, 25 MB size cap).
- Folder-based organisation: create, browse, and delete named folders per user.
- Full-text filename search within any folder.
- Secure per-user PDF download.
- Image asset management with categorized galleries (JPEG, PNG, GIF, WEBP, BMP).
- Free-tier plan with 3 uploads/day; upgradeable to an unlimited Pro plan.
- Responsive single-page application (SPA) frontend with real-time feedback.

3.2 Technology Stack

Layer	Technology	Purpose
Frontend	React 18 + Vite 6	Single-page application UI
UI Icons	Lucide-React	Consistent icon library
Backend	Python 3.13 + FastAPI 0.153	REST API server

ORM	SQLAlchemy 2.1	Database abstraction layer
Database	SQLite (via SQLAlchemy)	Relational data persistence
Auth	JWT (python-jose / passlib)	Stateless authentication
File Server	Uvicorn 0.54	ASGI application server
Settings	Pydantic-Settings 2.16	Environment-based config

4. Project Scope

4.1 In Scope

- User account management (register, login, logout, session expiry).
- PDF document upload, listing, search, download, and delete.
- Folder creation, listing, and deletion (with orphan-document handling).
- Image asset upload, listing, and deletion within user-defined categories.
- Tiered plan system (Free / Pro) enforced on the client side.
- Interactive API documentation via FastAPI's built-in Swagger UI.

4.2 Out of Scope

- PDF rendering or in-browser preview (documents are download-only).
- Real-time collaboration or multi-user shared folders.
- Payment processing or billing integration.
- Mobile native applications (iOS / Android).
- Cloud object-storage backends (e.g., AWS S3, Azure Blob).

5. Project Plan & Milestones

Phase	Milestone	Deliverable
1 – Planning	Requirements gathering & architecture design	SRS document
2 – Backend	FastAPI core, auth, document & image APIs	Working REST API
3 – Frontend	React SPA, Login, Dashboard, Upload panel	Functional UI
4 – Integration	End-to-end testing, CORS, env config	Integrated system
5 – Polish	Responsive CSS, error handling, plan gating	Production-ready build
6 – Docs	SPP, SRS, User Manual	Documentation package

6. Project Team

The project is developed and maintained by the EgoTECH engineering team. Roles include:

- Project Manager — oversees scope, timeline, and stakeholder communication.

- Backend Developer — FastAPI, SQLAlchemy, authentication, file handling.
- Frontend Developer — React, Vite, CSS, user experience design.
- QA Engineer — integration testing, API testing, bug triage.

7. Risk Analysis

Risk	Likelihood	Impact	Mitigation
Database corruption or data loss	Low	High	Regular SQLite backups; future migration to PostgreSQL
File-system storage fills up	Medium	Medium	Configurable storage path; plan quota enforcement
Unauthorized access to files	Low	High	JWT auth on all endpoints; per-user file isolation
Large file upload timeouts	Medium	Medium	25 MB server-side cap; streaming upload processing
Plan limit bypass by savvy users	Medium	Low	Move quota checks server-side in future release

8. Conclusion

EgoTECH PDF Vault is a practical, well-scoped document management solution built on modern, open-source technologies. It delivers immediate value to users who need a self-hosted alternative to heavy cloud storage platforms. This proposal outlines a clear path from development through delivery, supported by a detailed requirements specification and user manual.